

**AIMS**African Institute for
Mathematical Sciences
RWANDA

Introduction to Probability and Statistics

Lecture 5: Probability V

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Some classical discrete r.v.'s

1 Bernoulli

2 Binomiale

3 Poisson

Bernoulli's r.v.

Definition

A r.v. X is said to be a Bernoulli's r.v. with parameter p iff

$$P(X = x) = p^x(1 - p)^{1-x}, \quad x \in \{0, 1\}, \quad 0 < p < 1.$$

- **Notation:** $X \sim \mathcal{B}(1, p)$.
- **Expectation:** $E(X) = 0 \times (1 - p) + 1 \times p = p$.
- **Variance:**

Bernoulli's r.v.

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- **Expectation:** $E(X) = 0 \times (1 - p) + 1 \times p = p$.
- **Variance:**
 - $E(X^2) = 0^2 \times (1 - p) + 1^2 \times p = p$
 - $\text{Var}(X) = E(X^2) - (EX)^2 = p - p^2 = p(1 - p)$.

Binomiale distribution

Definition

A r.v. X follows a **binomiale distribution** with parameters (n, p) iff

$$P(X = x) = \binom{n}{x} p^x (1 - p)^{n-x}, \quad x \in \{0, \dots, n\} \quad 0 < p < 1.$$

• **Notation:** $X \sim \mathcal{B}(n, p)$

• $\sum_{x=0}^n P(X = x) = 1$ (Reminder: $\sum_{k=0}^n \binom{n}{k} a^k b^{n-k} = (a + b)^n$)

• **Exercise**

Use the moment generating function to find $E(X)$ and $\text{Var}(X)$.

Binomiale distribution

Solution:

$$\begin{aligned}M_x(t) &= E(e^{tX}) = \sum_{x=0}^n e^{tx} \binom{n}{x} p^x (1-p)^{n-x} \\&= \sum_{x=0}^n \binom{n}{x} (pe^t)^x (1-p)^{n-x} \\&= [(1-p) + pe^t]^n \text{ (applying the Newton's formula).}\end{aligned}$$

$$M'_X(t) = n[(1-p) + pe^t]^{n-1} \times pe^t; \quad M'_X(0) = E(X) = np.$$

$$M''_X(t) = n(n-1)[(1-p) + pe^t]^{n-2} (pe^t)^2 + n[(1-p) + pe^t]^{n-1} pe^t.$$

$$M''_X(0) = E(X^2) = n(n-1)p^2 + np = np[np + (1-p)].$$

$$\text{Var}(X) = E(X^2) - (EX)^2 = (np)^2 + np(1-p) - (np)^2 = np(1-p).$$

Moments generating function

Use the moment generating function to prove the following proposition:

Proposition

Let $X_1, X_2, \dots, X_n$ be n independent Bernoulli r.v. with parameter p . Then, $X = \sum_{i=1}^n X_i$ has a binomial distribution with parameter (n, p) .

Proof:

Moments generating function

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Proof:

We compute the m.g.f. of X_i :

$$\begin{aligned} M_{X_i}(t) &= E(e^{tX_i}) = e^{t \times 0} P(X = 0) + e^{t \times 1} P(X = 1) \\ &= (1 - p) + pe^t. \end{aligned}$$

From the proposition of the previous lecture:

$$M_X(t) = \prod_{i=1}^n M_{X_i}(t) = [(1 - p) + pe^t]^n.$$

We recognize the m.g.f of a binomiale r.v. with parameter (n, p) .

Expectation and Variance

Expectation and Variance

- **Expectation:**

Expectation and Variance

- **Expectation:** $E(X) = E\left(\sum_{i=1}^n X_i\right) = \sum_{i=1}^n E(X_i) = np.$
- **Variance:**

Expectation and Variance

- **Expectation:** $E(X) = E\left(\sum_{i=1}^n X_i\right) = \sum_{i=1}^n E(X_i) = np.$

- **Variance:** $Var(X) = Var\left(\sum_{i=1}^n X_i\right)$

$$= E\left[\left(\sum_{i=1}^n X_i - E\left(\sum_{i=1}^n X_i\right)\right)^2\right] = E\left[\left(\sum_{i=1}^n (X_i - EX_i)\right)^2\right]$$

$$= E\left[\sum_{i=1}^n (X_i - EX_i)^2 + \sum_{i \neq j} (X_i - EX_i)(X_j - EX_j)\right]$$

$$= \sum_{i=1}^n E[(X_i - EX_i)^2] + \sum_{i \neq j} E[(X_i - EX_i)(X_j - EX_j)]$$

$$= \sum_{i=1}^n Var(X_i) + \sum_{i \neq j} Cov(X_i, X_j)$$

Expectation and Variance

The X_i 's are independant, then $\text{Cov}(X_i, X_j) = 0$, for all $i \neq j$.

$$\text{Var}(X) = \sum_{i=1}^n \text{Var}(X_i) = \sum_{i=1}^n p(1-p) = np(1-p).$$

Computations using the definition

$$\begin{aligned} E(X) &= \sum_{x=0}^n xP(X=x) = \sum_{x=0}^n x C_n^x p^x (1-p)^{n-x} = \sum_{x=1}^n \frac{n!}{(x-1)!(n-x)!} p^x (1-p)^{n-x} \\ &= \sum_{x=1}^n \frac{n(n-1)!}{(x-1)!(n-x)!} p \times p^{x-1} (1-p)^{n-x}, \text{ let } k = x-1, \\ &= np \underbrace{\sum_{k=0}^{n-1} \frac{(n-1)!}{k!(n-1-k)!} p^k (1-p)^{n-1-k}}_{=1 \text{ Newton's formula}} \end{aligned}$$

Expectation and Variance

Variance : moment of order 2

$$\begin{aligned}
 E(X^2) &= \sum_{x=0}^n x^2 P(X=x) = \sum_{x=0}^n x^2 C_n^x p^x (1-p)^{n-x} = \sum_{x=1}^n x^2 \frac{n!}{x!(n-x)!} p^x (1-p)^{n-x} \\
 &= \sum_{x=1}^n x \frac{n(n-1)!}{(x-1)!(n-1-(x-1))!} p \times p^{x-1} (1-p)^{n-x}, \text{ let set } k = x-1, \\
 &= np \sum_{k=0}^{n-1} (k+1) \binom{n-1}{k} p^k (1-p)^{n-1-k} \\
 &= np \underbrace{\sum_{k=0}^{n-1} k \binom{n-1}{k} p^k (1-p)^{n-1-k}}_{=(n-1)p} + \underbrace{\sum_{k=0}^{n-1} \binom{n-1}{k} p^k (1-p)^{n-1-k}}_{=1 \text{ binôme de Newton}} \\
 &= np[(n-1)p + 1]
 \end{aligned}$$

$$\text{Hence } \text{Var}(X) = np[(n-1)p + 1] - (np)^2 = -np^2 + np = np(1-p).$$

Binomiale distribution

Exercise

Let $X \sim \mathcal{B}(m, p)$ and $Y \sim \mathcal{B}(n, p)$ with X and Y independent r.v.'s and $m \geq n$.

- ➊ Show that $Z = X + Y \sim \mathcal{B}(m + n, p)$.
- ➋ Find the conditional distribution of X given Z .

Solution:

➡ The proof of 1. is direct using the m.g.f.

Binomiale distribution

1

$$\begin{aligned}P(Z = z) &= P(X + Y = z) = \sum_{x=0}^z P(X = x, Y = z - x), \quad z = 0, 1, 2, \dots, m + n \\&= \sum_{x=0}^z P(X = x)P(Y = z - x) \text{ since } X \perp\!\!\!\perp Y \\&= \sum_{x=0}^z \binom{m}{x} p^x (1 - p)^{m-x} \binom{n}{z-x} p^{z-x} (1 - p)^{n-z+x} \\&= \sum_{x=0}^z \binom{m}{x} \binom{n}{z-x} p^z (1 - p)^{m+n-z}\end{aligned}$$

From the Vandermonde's formula $\sum_{x=0}^z \binom{m}{x} \binom{n}{z-x} = \binom{m+n}{z}$.

Therefore $P(Z = z) = \binom{m+n}{z} p^z (1 - p)^{m+n-z}$, $z = 0, 1, \dots, m + n$ which is the binomiale distribution with parameters $(m + n, p)$.

Binomiale distribution

Prove of the Vandermonde's formula: $\sum_{k=0}^{a+b} \binom{a}{i} \binom{b}{k-j} = \binom{a+b}{k}.$

From the Newton's formula :

$$(1+x)^{a+b} = \sum_{k=0}^{a+b} \binom{a+b}{k} x^k \quad (1)$$

But $(1+x)^{a+b} = (1+x)^a (1+x)^b.$

$$(1+x)^a (1+x)^b = \sum_{i=0}^a \binom{a}{i} x^i \sum_{j=0}^b \binom{b}{j} x^j = \sum_{i=0}^a \sum_{j=0}^b \binom{a}{i} \binom{b}{j} x^{i+j}.$$

Let $k = i + j$. k goes from 0 to $a + b$ and $j = k - i$ where i goes from 0 to k .
Therefore

$$(1+x)^a (1+x)^b = \sum_{k=0}^{a+b} \sum_{\{(i,j) | k=i+j\}} \binom{a}{i} \binom{b}{j} x^{i+j} = \sum_{k=0}^{a+b} \left[\sum_{i=0}^k \binom{a}{i} \binom{b}{k-i} \right] x^k$$

Then we deduce from (1) that $\sum_{i=0}^k \binom{a}{i} \binom{b}{k-i} = \binom{a+b}{k}.$

Binomiale distribution

② We find:

$$P(X = x \mid Z = z) = \frac{\binom{n}{x} \binom{m}{z-x}}{\binom{n+m}{z}}.$$

This is the **hypergeometric distribution**.

➡ This model arises in many situations.

For exemple, suppose an urn that contains n red balls and m white balls. A sample of size z is drawn from this urn (without replacement), then the number of red balls amongst the sample follows the hypergeometric distribution.

Poisson distribution

Definition

A r.v. X follow a **Poisson distribution** with parameter $\lambda > 0$ iff

$$P(X = x) = \frac{\lambda^x}{x!} e^{-\lambda}, \quad x \in \mathbb{N}.$$

- **Notation:** $X \sim \mathcal{P}(\lambda)$
- $\sum_{x \in \mathbb{N}} P(X = x) = 1$ (Recall: $e^x = \sum_{n=0}^{+\infty} \frac{x^n}{n!}$, $\forall x$.)
- **Exercise:** Compute $E(X)$ and $\text{Var}(X)$.

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- **Exercise:** Compute $E(X)$ and $Var(X)$.

$$EX = Var(X) = \lambda.$$

Poisson distribution

Solution:

$$\begin{aligned} E(X) &= \sum_{x=0}^{+\infty} xP(X=x) = \sum_{x=0}^{+\infty} x \frac{\lambda^x}{x!} e^{-\lambda} \\ &= \sum_{x=1}^{+\infty} \frac{\lambda^{x-1+1}}{(x-1)!} e^{-\lambda} = \lambda \sum_{x=1}^{+\infty} \frac{\lambda^{x-1}}{(x-1)!} e^{-\lambda} = \lambda \text{ since } \sum_{n=0}^{+\infty} \frac{u^n}{n!} = e^u. \end{aligned}$$

$$\text{Var}(X) = E(X^2) - (EX)^2.$$

$$\begin{aligned} E(X^2) &= \sum_{x=0}^{+\infty} x^2 P(X=x) = \sum_{x=0}^{+\infty} x^2 \frac{\lambda^x}{x!} e^{-\lambda} = \sum_{x=1}^{+\infty} x \frac{\lambda^{x-1+1}}{(x-1)!} e^{-\lambda} \\ &= \lambda \sum_{x=1}^{+\infty} (x-1+1) \frac{\lambda^{x-1}}{(x-1)!} e^{-\lambda} \\ &= \lambda \sum_{x=1}^{+\infty} (x-1) \frac{\lambda^{x-1}}{(x-1)!} e^{-\lambda} + \lambda \sum_{x=1}^{+\infty} \frac{\lambda^{x-1}}{(x-1)!} e^{-\lambda} = \lambda \\ &= \lambda E(X) + \lambda e^\lambda e^{-\lambda} = \lambda^2 + \lambda. \end{aligned}$$

Hence $\text{Var}(X) = \lambda$.

Poisson distribution

Exercise

Let X and Y be two independent r.v. X has a Poisson distribution with parameter λ_1 and Y has a Poisson distribution with parameter λ_2 .

- 1 Show that $X + Y$ has a Poisson distribution with parameter $\lambda_1 + \lambda_2$.
- 2 Show that the conditional distribution of X given $X + Y$ is a binomiale distribution.

Exercises

Exercise

Let us consider $X_1, X_2, \dots, X_n$, n v.a. independent identically distributed from a Poisson distribution with parameter λ .

Find the distribution of $Y = X_1 + X_2 + \dots + X_n$.